SCHOOL OF ENGINEERING AND TECHNOLOGY

ASSIGNMENT / PROJECT SUBMISSION FORM

PROGRAMME: BSc (Hons) Information Systems (Data Analytics)

SEMESTER: Jan / Apr / Aug 2023

SUBJECT: STAT1114 Statistics for Data Analytics

DEADLINE: 24th July 2023

INSTRUCTIONS TO CANDIDATES

- This is an ~~individual~~ / group project.

IMPORTANT
The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.

- Coursework submitted after the deadline but within 1 week will be accepted for a maximum mark of 40%.
- Work handed in following the extension of 1 week after the original deadline will be regarded as a non-submission and marked zero.

Lecturer's Remark (Use additional sheet if required)
List down the name of the group members and the student IDs here.

I Isabelle Yau Li Shuang (23021165), Leow Yan Yi (23020324), Peggy Liow En Ki (23020852), Yeo Khaining (23033152) received the assignment and read the comments.

Isabelle Yau, Leow Yan Yi, Peggy Liow, Yeo Khaining (26/07/2023)

Academic Honesty Acknowledgement
"I Isabelle Yau Li Shuang, Leow Yan Yi, Peggy Liow En Ki, Yeo Khaining (Student's Name) verify that this paper contains entirely my own work. I have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, I have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. I realize the penalties *(refer to page 16, 5.5, Appendix 2, page 44 of the student handbook diploma and undergraduate programme)* for any kind of copying or collaboration on any assignment."

Isabelle Yau, Leow Yan Yi, Peggy Liow, Yeo Khaining (26/07/2023)

Data Protection
The protection of personal data is an important concern to Sunway University and any personal data collected on this form will be treated in accordance with the Personal Data Protection Notice of the institution.
http://sunway.edu.my/pdpa/notice_english (English version)
http://sunway.edu.my/pdpa/notice_bm (Malay version)

1. Introduction

This paper focuses on the price per unit area of houses and the factors that may affect the price. Since there are numerous variables that may affect house prices, this paper compares and explores the most influential factor.

These are some objectives that this paper focuses on:

- Construct a Mallow's cp plot to explore the relationship between a response variable and multiple predictor variables.
- Build a multiple linear regression model (using model Mallow's cp methods)
- Use a multiple linear regression model for prediction and analytical or explanatory analysis.

In addition, these are some variables this paper may use to explore factors that may influence the price:

- house_age

"house_age" interpret the age of the house in years.

- distance_MRT

"distance_MRT" defines the distance from the house to the nearest MRT station in metres.

- distance_school

"distance_school" defines the distance from the school to the house in metres.

- number_stores

"number_stores" defines the number of convenience or grocery stores within a 3km radius of the house.

- bedrooms

"bedrooms" defines the number of bedrooms in the house.

- bathroom

"bathroom" defines the number of bathrooms in the house.

- renovated

"renovated" defines whether the house has been renovated. This paper uses 0 and 1 to describe the status of renovated houses where '0' denotes that the house has not been renovated and '1' indicates that it has.

- price_per_unit

"price_per_unit" defines the price per unit area of the house.

- sold_3months.

"sold_3months" is an indicator variable on whether the house is sold within three months. This paper uses '0' to denote houses which are not sold within three months and '1' indicates that it has.

- sold_6months.

"sold_6months" is another indicator variable on whether the house is sold within six months. This paper uses '0' to denote houses which are not sold within six months and '1' indicates that it has.

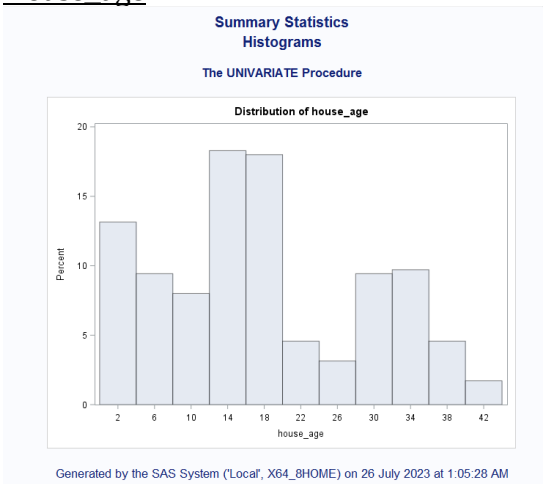
2. Descriptive Analysis

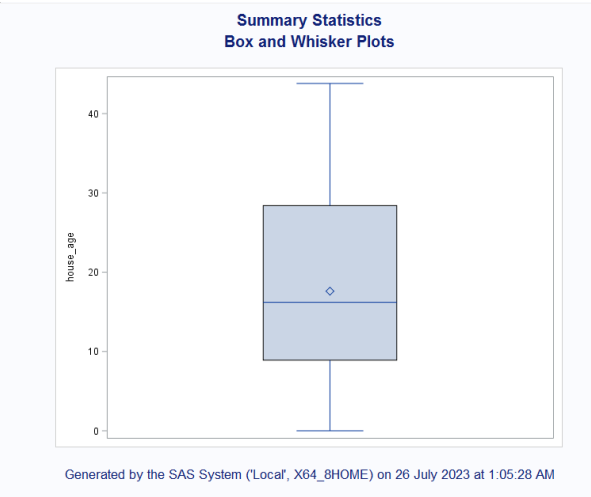
Summary Statistics Results

The MEANS Procedure

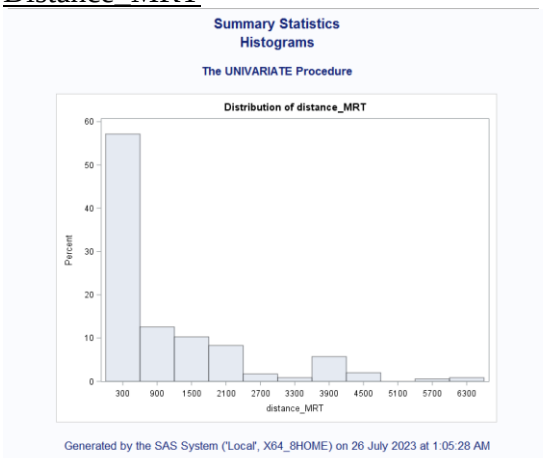
Variable	Mean	Std Dev	Minimum	Maximum	N
house_age	17.6040000	11.3679530	0	43.8000000	350
distance_MRT	1095.76	1283.52	23.3828400	6488.02	350
distance_school	1086.19	1283.40	13.3847823	6469.93	350
number_stores	4.0542857	2.9542687	0	10.0000000	350
bedrooms	3.0771429	1.4110870	1.0000000	5.0000000	350
bathrooms	1.9142857	0.8072532	1.0000000	3.0000000	350
renovated	0.4857143	0.5005114	0	1.0000000	350
sold_3months	0.5028571	0.5007076	0	1.0000000	350
sold_6months	0.7171429	0.4510324	0	1.0000000	350
price_per_unit_area	37.4862857	12.9829063	11.2000000	78.3000000	350

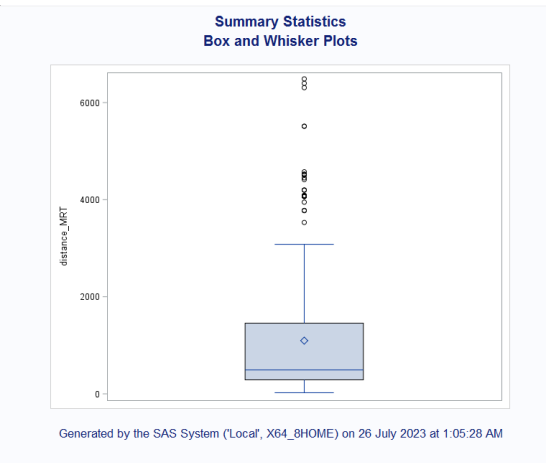
House_age



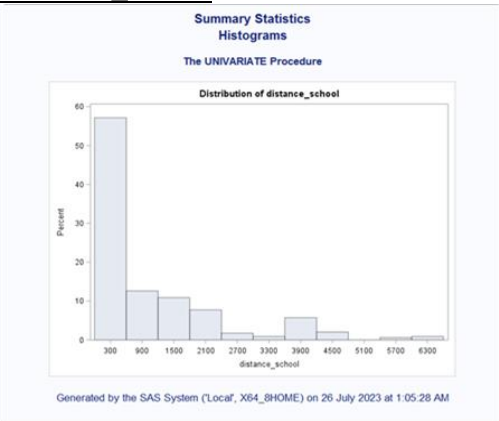


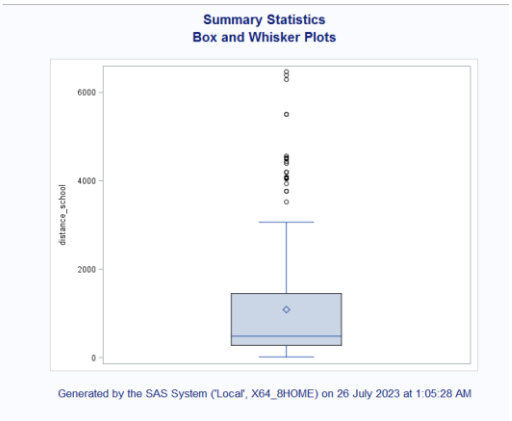
Distance_MRT



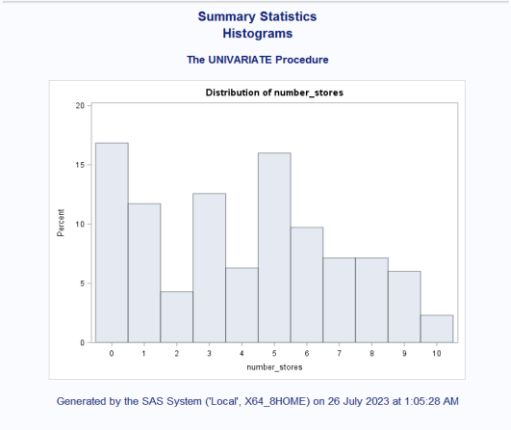


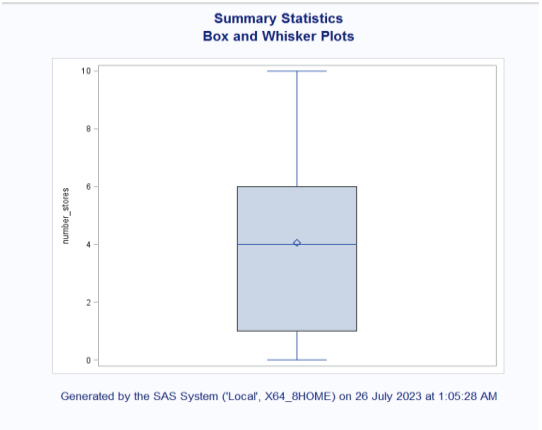
Distance_school



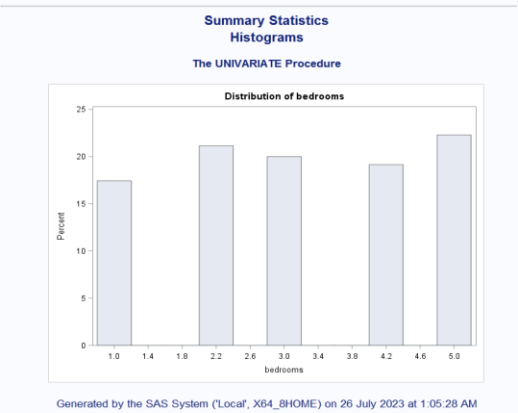


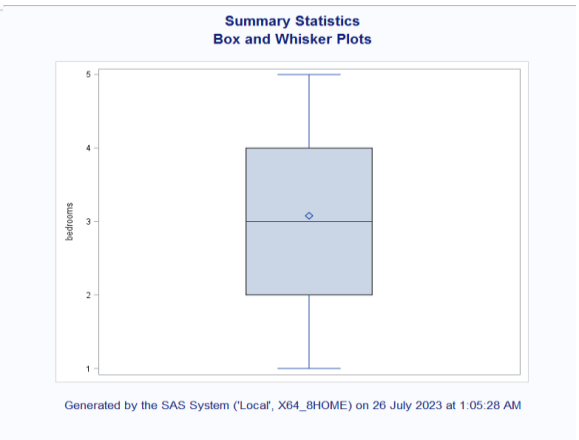
Number_stores



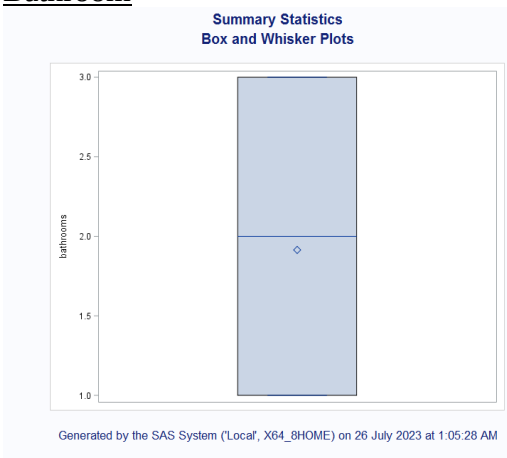


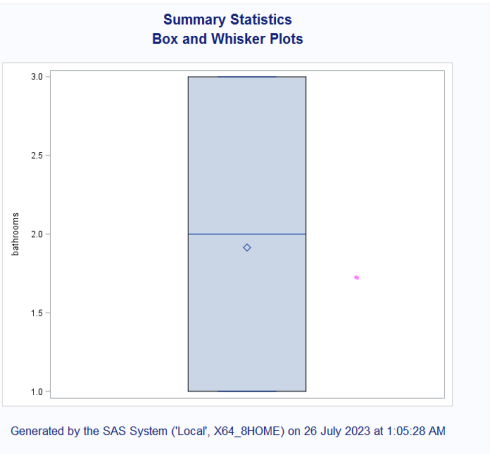
Bedrooms



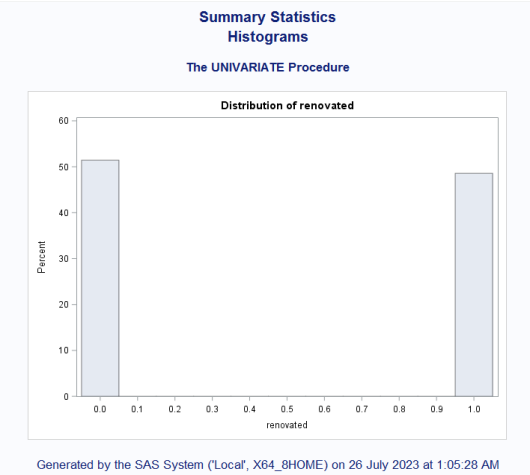


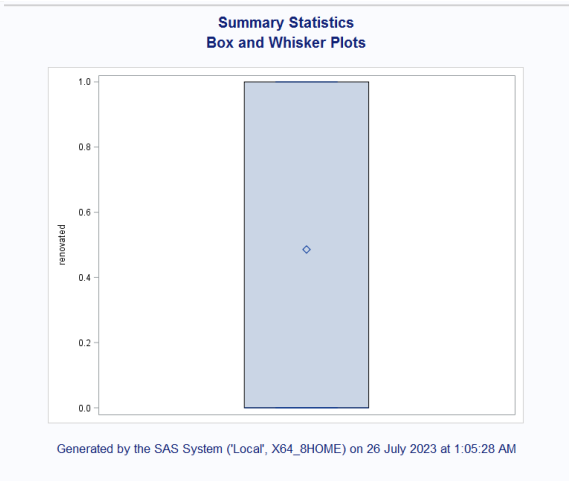
Bathroom



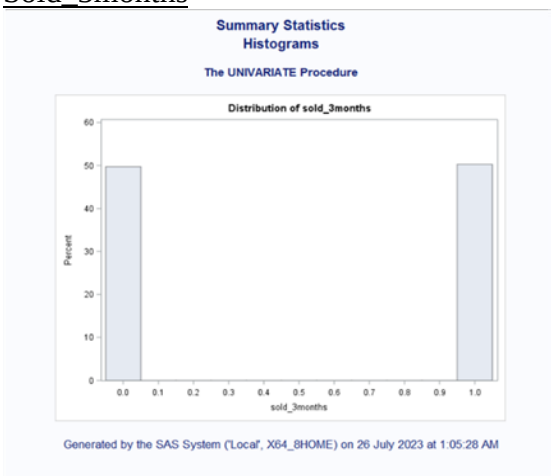


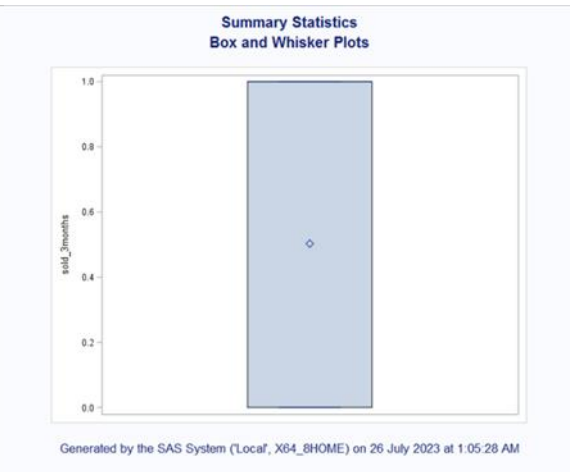
Renovated



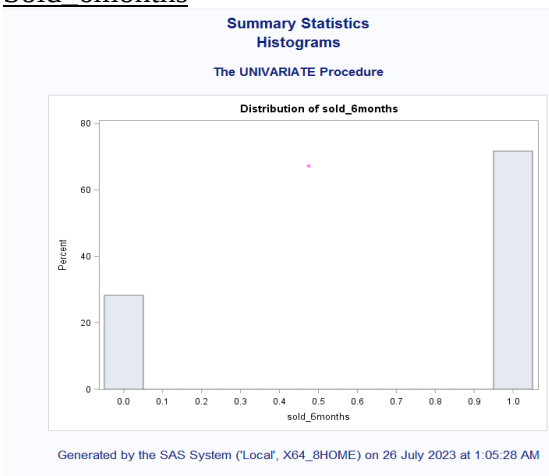


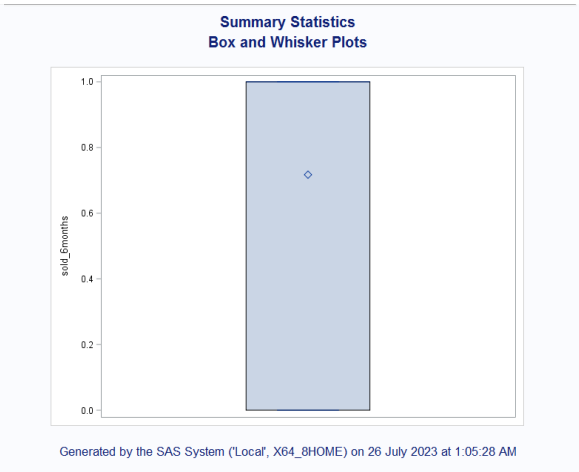
Sold_3months



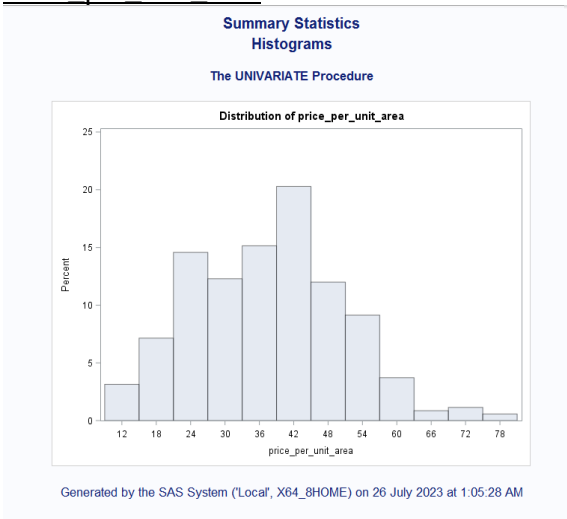


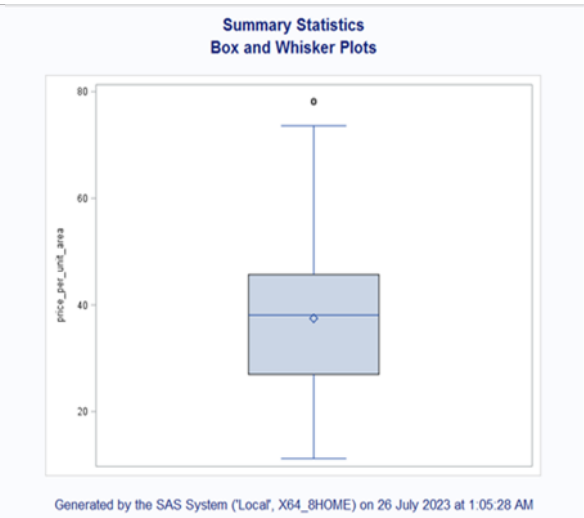
Sold_6months

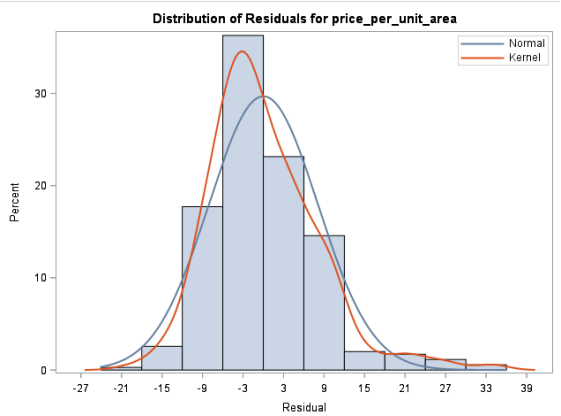


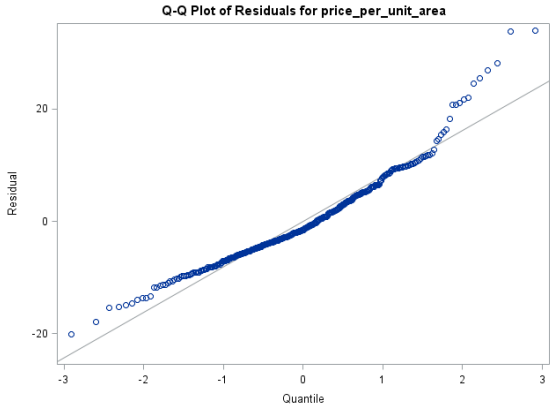


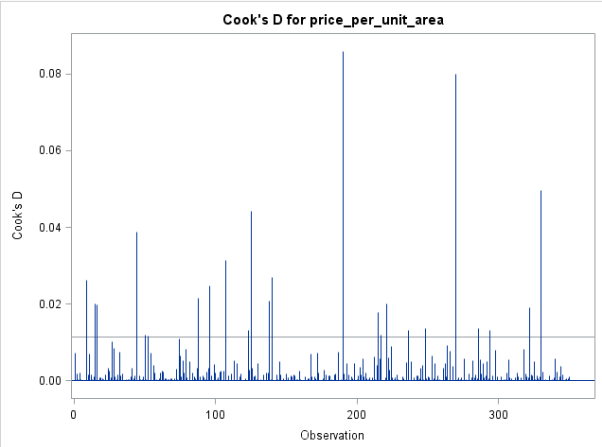
Price_per_unit_area











3. Statistical Inference and Modelling

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: price_per_unit_area

Number of Observations Read	350
Number of Observations Used	350

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	36138	12046	183.71	<.0001
Error	346	22688	65.57134		
Corrected Total	349	58826			

Root MSE	8.09761	R-Square	0.6143
Dependent Mean	37.48629	Adj R-Sq	0.6110
Coeff Var	21.60154		

Parameter Estimates								
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Standardized Estimate	95% Confidence Limits	
Intercept	1	41.50166	1.30788	31.73	<.0001	0	38.92927	44.07405
house_age	1	-0.26097	0.03822	-6.83	<.0001	-0.22850	-0.33615	-0.18579
distance_MRT	1	-0.00503	0.00042078	-11.96	<.0001	-0.49746	-0.00586	-0.00420
number_stores	1	1.50271	0.18302	8.21	<.0001	0.34194	1.14274	1.86267

H0: β (price_per_unit_area) = β (house_age) = β (distance_MRT) = β (distance_school) = β (number_stores) = β (bedrooms) = β (bathrooms) = β (renovated) = β (sold_3months) = β (sold_6months) = 0
H1: At least one $\beta \neq 0$.

The F-value is 183.71, with corresponding p-value < 0.0001.
The p-value is < 0.0001, which is smaller than the significance value (α = 0.05).
Hence, H0 is rejected.
At least one of the explanatory variables have a significant effect on the response variable, price per unit area.

The adjusted R-Square value for the multiple regression model is 0.6110.
Therefore, 61.10% of the variation in price per unit area is explained by the multiple regression model and is adjusted for the number of independent variables and sample size.

The standard error of estimate is reported by Root MSE = 8.09761.
The typical difference between actual price per unit area and the predicted price per unit area using the regression equation is approximately $8.09761.

The regression equation for these data is $\hat{Y}$= 41.5017-0.2610x_1-0.0050x_2+1.5027x_3.
The equation describes the effect of an independent variable controlling effects of the other independent variables in the model.

-0.2607 is interpreted as the change in price per unit area of houses (Y) corresponding to a unit change in house age (X1) when distance of MRT (X2) and number stores (X3) are held constant.

When the distance of MRT (X2) increase by one metre, the price per unit area of houses (Y) decrease by 0.0050, when house age (X1) and number stores (X3) are held constant.

When the number of stores (X3) increases by one unit, the price per unit area of houses (Y) increases by 1.5027, when house age (X1) and distance of MRT (X2) are held constant.

House_age

Parameter Estimates								
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Standardized Estimate	95% Confidence Limits	
Intercept	1	41.50166	1.30788	31.73	<.0001	0	38.92927	44.07405
house_age	1	-0.26097	0.03822	-6.83	<.0001	-0.22850	-0.33615	-0.18579
distance_MRT	1	-0.00503	0.00042078	-11.96	<.0001	-0.49746	-0.00586	-0.00420
number_stores	1	1.50271	0.18302	8.21	<.0001	0.34194	1.14274	1.86267

Generated by the SAS System ('Local', X64_8HOME) on 23 July 2023 at 9:12:26 PM

$H0$: β(house_age) = 0
$H1$: β(house_age) $\neq$ 0
The test statistics t-value = -6.83 with corresponding p-value is <0.0001,
H0 is rejected at significance level α = 0.05.
There is strong evidence that house age is related to the price per unit area, controlling for distance MRT and number of stores.
Controlling for other independent variables in the model, we are 95% confident that the change in mean price per unit area increase in house age falls between -0.33615 and -0.18579.

Distance_MRT

Parameter Estimates								
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Standardized Estimate	95% Confidence Limits	
Intercept	1	41.50166	1.30788	31.73	<.0001	0	38.92927	44.07405
house_age	1	-0.26097	0.03822	-6.83	<.0001	-0.22850	-0.33615	-0.18579
distance_MRT	1	-0.00503	0.00042078	-11.96	<.0001	-0.49746	-0.00586	-0.00420
number_stores	1	1.50271	0.18302	8.21	<.0001	0.34194	1.14274	1.86267

Generated by the SAS System ('Local', X64_8HOME) on 23 July 2023 at 9:12:26 PM

$H0$: β(distance_MRT) = 0
$H1$: β(distance_MRT) $\neq$ 0
The test statistic t Value= -11.96 with corresponding p-value is <0.0001,
H0 is rejected at significance level α=0.05.

There is strong evidence that distance MRT is related to the price per unit area, controlling for house_age and number of stores.

Controlling for other independent variables in the model, we are 95% confident that the change in mean price per unit area increase in distance MRT falls between -0.00586 and -0.00420.

Number_stores

Parameter Estimates								
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Standardized Estimate	95% Confidence Limits	
Intercept	1	41.50166	1.30788	31.73	<.0001	0	38.92927	44.07405
house_age	1	-0.26097	0.03822	-6.83	<.0001	-0.22850	-0.33615	-0.18579
distance_MRT	1	-0.00503	0.00042078	-11.96	<.0001	-0.49746	-0.00586	-0.00420
number_stores	1	1.50271	0.18302	8.21	<.0001	0.34194	1.14274	1.86267

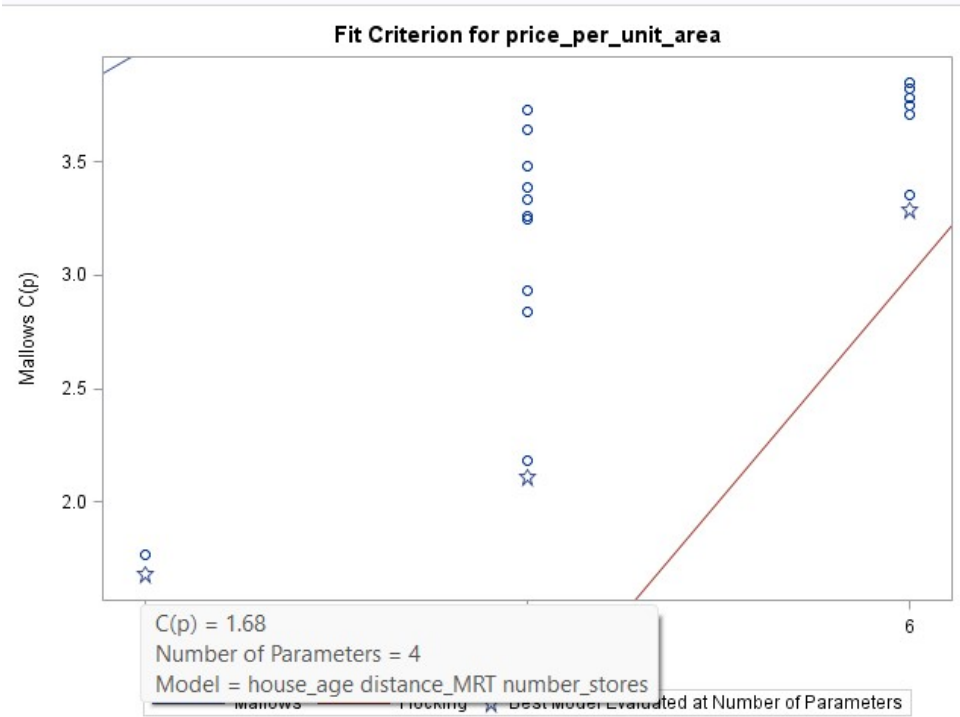
Generated by the SAS System ('Local', X64_8HOME) on 23 July 2023 at 9:12:26 PM

$H0$: β (number of stores) = 0
$H1$: β (number of stores) $\neq$ 0
From the SAS output, t value=8.21 with corresponding p-value <0.0001.

The p-value is < 0.0001, thus H0 is rejected at the 0.05 level of significance (α=0.05).

There is strong evidence that the number of stores is related to the price per unit area, controlling for house_age and distance of MRT.

Controlling for other independent variables in the model, we are 95% confident that the change in mean price per unit area increase in distance MRT falls between 1.14274 and 1.86267.



The model with house age, distance MRT and number of stores as the main variables is chosen based on Mallow's Cp criteria. This model is chosen because it's C(p) value is the smallest (1.68), at it is located below the line which is indicated by the star.

4. Conclusion

After constructing the Mallow's Cp plot, we could explore the relationship between the price per unit area which is the response variable and the multiple predictor variables which are house age, distance MRT, distance school, number stores, bedrooms, bathroom, renovated, sold 3 months and sold 6 months. The null hypothesis of the model is rejected as the p-value is <0.0001, smaller than the significance value (0.05). At least one of the explanatory variables have a significant effect on the response variable. House age, number stores and distance MRT are the variables that show their significant effect towards price per unit area. As the house age increases, the condition of the house will be affected thus affecting the price per unit area. The distance of MRT affects price per unit area due to its convenience in terms of transport. Besides, the number of stores affects price per unit area as it provides convenience for the residents to buy daily necessities and other products.